

JeevaFlux - Complete System & Module Documentation

On-Device AI Personal Health Companion for early detection and early action

JeevaFlux • Team SYMBIONICS • Smart India Hackathon 2026 • SIH26181

1. Project Overview

JeevaFlux is proposed as a secure, AI-powered personal health companion for real-time, privacy-preserving health monitoring and early warning. It is designed around the interaction between a person's physiological state and environmental conditions, with on-device intelligence for low-latency operation and continued support when connectivity is unavailable.

Item	JeevaFlux
Problem Statement	SIH26181 -- secure, AI-powered Personal Health Companion
Theme	MedTech / BioTech / HealthTech
Category	Hardware
Team	SYMBIONICS
Primary design principle	On-device intelligence + privacy-preserving monitoring
Operating model	Wearable/device + smartphone + optional cloud synchronization



2. Problem Addressed

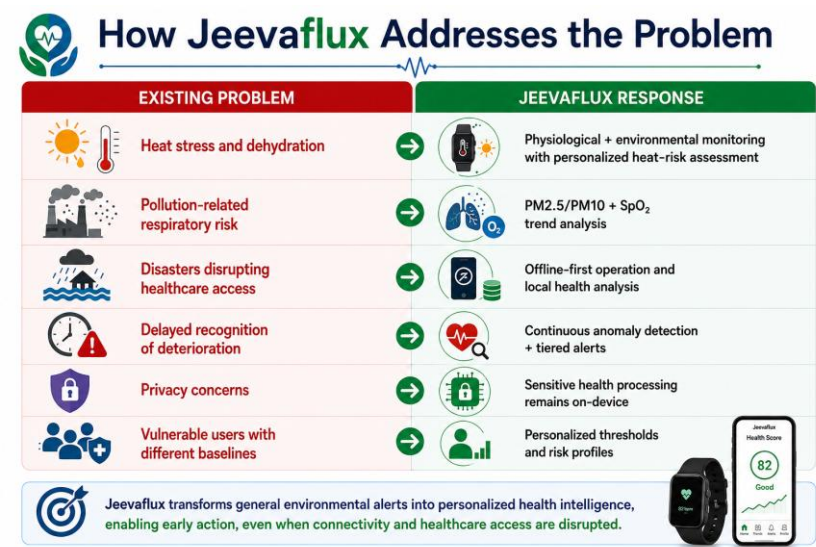
- Delayed recognition of health deterioration can turn an early warning into an emergency.
- Heat waves, floods, pollution events and other disasters can expose individuals to rapidly changing physiological and environmental risks.
- Continuous monitoring can be difficult when connectivity is intermittent or absent.
- Cloud-only processing raises latency, availability and privacy concerns for sensitive health information.

- A useful system must interpret physiological signals together with environmental exposure and individual context rather than treating each measurement independently.



3. Proposed Solution

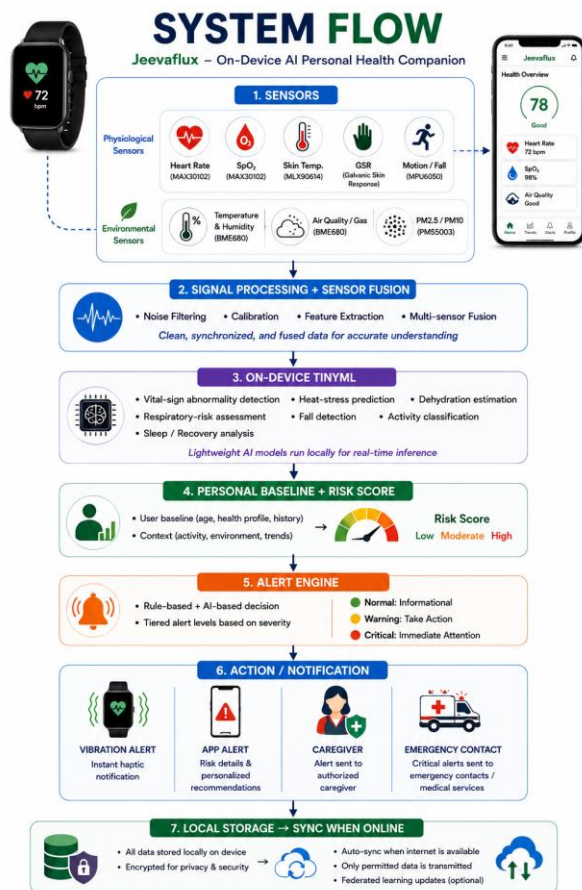
- Continuous acquisition of physiological and environmental measurements through a wearable/edge device.
- Local sensor fusion and lightweight AI inference to identify abnormal or deteriorating states.
- Personalized early warnings based on an individual's baseline and relevant context.
- Mobile companion for visualization, alerts, guidance and emergency-oriented actions.
- Offline-first operation with local processing, local storage and store-and-forward synchronization when connectivity returns.
- Privacy-by-design architecture that minimizes transmission of raw sensitive data.



4. Core Architecture

Layer	Responsibilities
Sensing Layer	Collect physiological and environmental signals from

	wearable/edge sensors.
Edge Intelligence Layer	Filtering, calibration, anomaly detection, feature extraction and lightweight AI inference.
Risk & Context Layer	Combine physiological state, environmental exposure and personal context into a risk assessment.
Alert Layer	Generate graded warnings and actionable guidance rather than raw sensor notifications.
Mobile Companion	Display current status, trends, warnings, guidance and user-facing controls.
Synchronization Layer	Securely synchronize selected records when connectivity becomes available.
Analytics / Institutional Layer	Optional aggregated analytics for authorized deployments; exact scope to be defined during pilot.



5. Key Inputs

- Physiological: heart rate, SpO₂, skin temperature, activity/motion, sleep/recovery indicators and GSR where supported by the device.
- Environmental: temperature, humidity, particulate pollution such as PM2.5/PM10 and air-quality indicators.
- Context: personal baseline, activity level, exposure duration and user-selected vulnerability information.
- Location/time context may be used where explicitly enabled for disaster or environmental risk awareness.

6. Early-Warning Modules

Module	Function
Heat / Heat-Stress Risk	Combine temperature/humidity exposure with physiological response and activity to identify rising heat burden and provide early warnings.
Dehydration / Physiological Strain	Use changes in physiological signals together with exposure/activity context to flag possible strain and prompt preventive action. This is an early-warning aid, not a diagnosis.
Respiratory / Pollution Risk	Relate air-quality exposure to physiological/context signals and surface elevated respiratory-risk conditions.
Abnormal Vital-Sign / Anomaly Detection	Continuously compare measurements with personalized baseline and detect meaningful deviations.
Fatigue / Poor Recovery	Use activity and sleep/recovery patterns to identify sustained fatigue or inadequate recovery.
Fall / Distress Support	Use motion patterns and available contextual signals to identify possible falls or distress events and initiate an appropriate alert workflow.

7. Personalized Risk Logic

1. Acquire and validate incoming sensor measurements.
2. Remove obvious noise and reject unreliable readings where possible.
3. Extract short-window and trend features rather than relying only on a single measurement.
4. Compare features against the individual's baseline and contextual conditions.
5. Fuse multiple signals to reduce dependence on one noisy sensor.
6. Assign a graded risk state such as normal, caution, high risk or emergency candidate.
7. Deliver an understandable action-oriented warning and log the event for later review.

8. Mobile Companion

- Personal health status and recent trends.
- Environmental exposure/risk information.
- Early-warning notifications with clear recommended actions.
- History and event timeline.
- Device/connectivity/battery status.
- Emergency-oriented controls and contact workflows as defined during implementation.
- Regional-language support and assisted onboarding are proposed adoption features.

9. Offline-First Operation

- Core monitoring and early-warning inference should continue locally without continuous internet.
- Events and selected time-series summaries can be stored locally.
- When connectivity returns, the system can synchronize permitted records using a store-and-forward mechanism.
- Cloud dependence should not be required for time-critical local warning generation.

10. Privacy & Security

- Process sensitive health information locally wherever practical.
- Collect and transmit only the data necessary for the enabled feature.

- Use explicit consent for optional sharing.
- Protect synchronized records through authenticated and encrypted communication.
- Maintain auditable access and retention policies for any institutional deployment.

11. User Groups

- Elderly citizens and other users who benefit from continuous early-warning support.
- Outdoor workers exposed to heat and environmental stress.
- Individuals vulnerable to pollution or respiratory stress.
- People living or working in low-connectivity areas.
- Disaster-prone communities requiring resilient local monitoring.

12. End-to-End Workflow

8. User wears/activates the JeevaFlux device.
9. Physiological and environmental sensors begin periodic acquisition.
10. Edge processing validates and filters the measurements.
11. Features are generated over short time windows and compared with personal baseline/context.
12. The risk engine performs multimodal sensor fusion.
13. If risk is detected, JeevaFlux generates a graded alert and recommended action.
14. The event is recorded locally; permitted data are synchronized later when connectivity is restored.
15. The user or authorized support system can review the event and response.

13. Prototype-to-Deployment Position

The SIH deck positions JeevaFlux as an on-device AI health companion and presents deployment, feasibility, risk-mitigation and measurable-impact framing. The exact clinical validation status, regulatory classification, final sensor bill of materials and field performance figures should be treated as validation items rather than established claims until experimentally demonstrated.